Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Results-driven Data Scientist and AI Solutions Engineer with 3 years of industry experience developing scalable GenAI applications, implementing RAG architectures, and driving data automation. Recently completed an MSc in Data Science at the University of Bristol, highlighted by an algorithmic dissertation project in partnership with AstraZeneca. A proven problem-solver with a track record of winning multiple UK hackathons. Seeking to leverage my expertise in machine learning, Python, and cloud infrastructure to deliver impactful, data-driven solutions in a dynamic UK-based organization.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Retrieval Augmented Generation (RAG) Engineer

- Developed **GenAI applications** using **LlamaIndex** and **Azure OpenAI GPT-3.5 & Llama-3** for internal company use to answer department-specific queries.
- Designed and Maintained **Vector Database** using **pgvector** and **OpenAI embeddings model** to enable efficient semantic search and retrieval across internal company documents.
- Built a **WhatsApp** fund-comparison bot (via the **Netcore** API) used daily by the Sales team (roughly **50 to 100 users**, **400 to 1,000 requests** in working hours): **Azure OpenAI tool calling** picks the funds and comparison type and calls **Python functions** over SQL data, returning fund indicators and a chart PDF. Parameter extraction broke first; fixed with example inputs in the **JSON tool definitions**.
- Developed a **Python** automation job utilizing **AWS Textract** to parse over **300 daily emails**, transforming unstructured data into a structured format and **saving 60 manual analyst hours a month**.

Data Analyst

- Analyzed customer data and drew insights using **Python Pandas**.
- Developed **Power BI** dashboards for fortnightly credit report analysis thus, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**.
- Identified customer buying patterns using **Market Basket Analysis** by implementing the **Apriori Algorithm**.

Data Engineer

- Designed **Normalized SQL Schema** for internal analytics platform.
- Developed **Optimized SQL** queries for data preprocessing.
- Automated Data preprocessing pipelines using **AWS SageMaker**.
- Developed **SQL Functions** and **Procedures** for efficient data insertion.

MLOps

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on AWS utilizing core services such as **EC2, S3, ECR and Lambda.**
- Administered Security roles and policies using **AWS Security Groups**.

- Deployed **Nginx** reverse proxy to minimize external security threat to below **1%**.
- Coordinated deployment automation of multiple environments by harnessing **Jenkins** and **GitLab** using a **CI/CD pipeline**.
- Managed multiple **Anaconda Python** environments using **Docker**.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Algorithm to rank best suited cell lines based on target inclusion and exclusion genes.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolution Neural Network (CNN)** model.)
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Multi-phase architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention/transformer decoders for generating natural-language image descriptions, benchmarked against BLIP/SigLIP and commercial VLMs.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Pauti:** Expense tracking and payment splitting Android application. (Uses an offline only, privacy first principle by implementing a **CRDT** architecture.) | GitHub: https://github.com/Arishtanemi4/pauti.git
- **Swindon Borough Council Fuel-Poverty Hack:** A household and neighborhood-based fuel poverty data analysis tool. (Made for the Swindon Borough Council Fuel Poverty Hackathon.) | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git
- **Contact Reminder:** A simple Android application to text contacts on their important dates like birthday or anniversary. (Implemented phone call and WhatsApp integration for seamless experience.) | GitHub: https://github.com/Arishtanemi4/contact-reminder.git

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS, AzureOpenAI, Power BI, Nginx, Git, Putty, Linux, VS Code, Docker, Jenkins, GitLab, Claude Code.
- **Languages:** Python (pandas, FastAPI, scikit-learn), R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL, SQLite